PROJECT REPORT

ON

Design and Implementation of a Secure Segmented Network using Inter-VLAN Routing and Access

Submitted to

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(Off-Campus Centre, Nitte Deemed to be University, Nitte - 574 110, Karnataka, India)

In partial fulfilment of the requirements for the award of the

Degree of Bachelor of Technology

in

COMPUTER SCIENCE AND ENGINEERING (CYBERSECURITY)

by

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2025 – 2026

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CERTIFICATE

This is to certify that **Ms. Shreelakshmi Prabhu (NNM24CB053)** and **Mr. Prithvik K S (NNM25CB504)** of II-year B.Tech., a bonafide students of NMAM Institute of Technology, Nitte, has carried out project on "**Design and Implementation of a Secure Segmented Network using Inter-VLAN Routing and Access**" as part of the **Unix System Programming and Networking Lab (CB2601-1)** course during 2025-26, fulfilling the partial requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering (Cyber Security) at NMAM Institute of Technology, Nitte.

.....

Name and Signature of Course Instructor

ABSTRACT

In contemporary network environments, the reliance on unsegmented, flat architectures presents a significant security risk, as it allows for unrestricted lateral movement of threats and unauthorized access to sensitive departmental data. Without proper logical boundaries, a single compromised device can jeopardize the entire infrastructure, leading to data breaches and unmanaged traffic congestion. This project addresses these vulnerabilities by adopting a "Defense in Depth" approach through network segmentation. By leveraging Virtual Local Area Networks (VLANs), the design focuses on isolating organizational units—specifically Staff, Guest, Admin, and Server zones—to ensure that traffic is strictly confined to its designated domain while maintaining necessary inter-departmental communication.

The solution was implemented using Cisco Packet Tracer, employing a Router-on-a-Stick configuration to enable controlled inter-VLAN routing through sub-interfaces and 802.1Q encapsulation. To harden the infrastructure, security measures were integrated at multiple layers, including Layer 2 Port Security to prevent unauthorized physical connections and Extended Access Control Lists (ACLs) to enforce the principle of least privilege for server access. Additionally, the management plane was secured using encrypted SSH v2 to protect against credential sniffing. The final outcome is a resilient and fully functional network where administrative control is centralized, guest access is restricted, and critical server resources are shielded from unauthorized probes. Verification through protocol analysis and ICMP testing confirms that the architecture successfully balances high-performance connectivity with robust security standards.

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CHAPTER 1

INTRODUCTION

1.1 Purpose of Network Segmentation

Network segmentation involves dividing a network into smaller, isolated zones to improve security and performance. In a "flat" network, a single breach can compromise every device. By using VLANs, this project achieves:

- Contain attacks by limiting lateral movement
- Reduce network congestion
- Enforce least privilege access

1.2 Project Scope and Security Objectives

This project covers the design of a secure LAN using Cisco Packet Tracer, focusing on core switching, routing, and infrastructure hardening.

Security Objectives:

- Isolate networks (Staff, Guest, Admin, Server)
- Prevent unauthorized access using port security
- Control traffic using ACLs
- Secure management using SSH

1.3 Technical Overview

The network follows a Star Topology using a Cisco 2960 Switch and a Cisco 2911 Router.

Key Technologies:

- Router-on-a-Stick (inter-VLAN routing)
- IPv4 addressing (192.168.x.0/24)
- Cisco IOS security (RSA, Sticky MAC)

CHAPTER 2

REQUIREMENT SPRECIFICATION

2.1 Hardware Specification

- Router: Cisco 2911 ISR
- Switch: Cisco 2960-24TT Managed Switch
- End Devices: 4 Generic PCs (Staff, Guest, Admin)
- Servers: 2 Generic Servers (Web and Database)
- Connectivity: Copper Straight-Through and Gigabit Ethernet cabling

2.2 Software Specification

The implementation and testing of the network design relied on the following software tools:

- Cisco Packet Tracer (v8.2 or higher): The primary simulation tool used for designing the topology, configuring Cisco IOS devices, and performing real-time packet analysis.
- Cisco IOS (Internetwork Operating System): The command-line interface (CLI) software running on the router and switch, used for executing security configurations ($SSH, ACLs, Port-Security$).
- Web Browser (Simulation): Used on end-user PCs to verify HTTP access to the Server VLAN.
- Command Prompt (Terminal): Used for troubleshooting and verifying connectivity through ping.
- Operating System: Windows 10/11 (Host machine) to run the simulation environment.

CHAPTER 3

SYSTEM DESIGN

3.1 Logical Network Topology

The logical network topology illustrates a segmented VLAN-based architecture where staff,guest,admin, and server networks are isolated and interconnected through a central switch and router to ensure secure and efficient communication

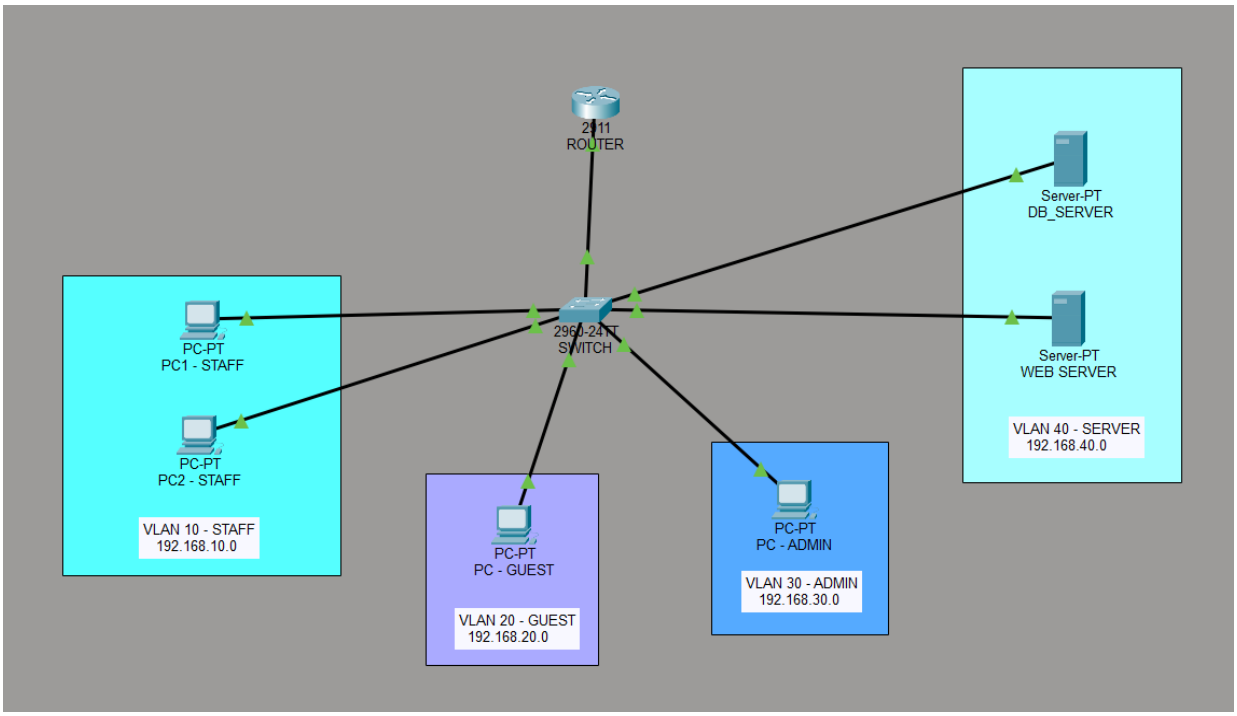


Figure 3.1: Logical Network Topology showing VLAN segmentation

3.2 VLAN Planning and subnetting

VLAN ID	Department	Network Address	Default Gateway	Device(s)
10	Staff	192.168.10.0/24	192.168.10.1	PC1, PC2
20	Guest	192.168.20.0/24	192.168.20.1	PC-GUEST
30	Admin	192.168.30.0/24	192.168.30.1	PC-ADMIN
40	Servers	192.168.40.0/24	192.168.40.1	Server-WEB SERVER Server-DB_SERVER

Table 3.1 VLAN Allocation and Addressing Scheme

3.3 Access Control and Security Policy Design

The security architecture is built on the Principle of Least Privilege (PoLP). The following policies were designed before implementation:

- Layer 2 Policy: Enable Port Security on all access ports to restrict access to "Sticky" MAC addresses only, preventing rogue device insertion.
- Layer 3 Policy (ACLs): * Staff: Permit HTTP access to Web Server; Deny ICMP (Ping) to Server VLAN.
    - Guest: Full isolation; Deny access to Staff and Server VLANs.
    - Admin: Unrestricted access to all VLANs for maintenance and monitoring.
- Management Policy: Disable Telnet; enforce SSH v2 for encrypted remote configuration, restricted to the Admin IP.

CHAPTER 4

IMPLEMENTATION

4.1 VLAN Creation and Trunking Configuration

The switch was configured to logically separate the departments. A trunk link was established on interface Fa0/24 to allow the router to identify traffic from different VLANs using the IEEE 802.1Q standard.

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name STAFF
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name GUEST
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name ADMIN
Switch(config-vlan)# exit
Switch(config)# vlan 40
Switch(config-vlan)# name SERVER
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# interface fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
Switch(config)# interface range fa0/5-6
```

```
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 40
Switch(config-if)# exit
Switch(config)# interface fa0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch(config)# exit
```

4.2 Inter-VLAN Routing (Router-On-A-Stick)

To allow communication between the segmented networks, the Cisco 2911 Router's physical interface (G0/0) was divided into logical sub-interfaces. Each sub-interface serves as the default gateway for its respective subnet.

```
Router> enable
Router# configure terminal
Router(config)# interface g0/0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface g0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface g0/0.40
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
Router(config-subif)# exit
```

4.3 Layer 2 Hardening (Port Security and MAC Filtering)

Port security was implemented on the switch to mitigate unauthorized physical access. By using the "Sticky" feature, the switch learns the legitimate MAC address of the connected PC and shuts down the port if a rogue device is detected.

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range fa0/1-6
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport port-security
Switch(config-if-range)# switchport port-security maximum 1
Switch(config-if-range)# switchport port-security mac-address sticky
Switch(config-if-range)# switchport port-security violation shutdown
Switch(config-if-range)# exit
Switch(config)# exit
```

4.4 Management Plane Security (SSH v2 and VTY Configuration)

To protect administrative access from eavesdropping, unencrypted Telnet was disabled. SSH v2 was enabled using RSA encryption, and access was restricted via an access-class to ensure only the Admin PC can manage the device.

```
Router> enable
Router# configure terminal
Router(config)# hostname Router
Router(config)# ip domain-name secure.local
Router(config)# crypto key generate rsa
1024
Router(config)# username admin secret cisco
Router(config)# access-list 10 permit 192.168.30.2
Router(config)# line vty 0 4
Router(config-line)# login local
Router(config-line)# transport input ssh
Router(config-line)# access-class 10 in
Router(config-line)# exit
Router(config)# ip ssh version 2
Router(config)# end
```

4.5 Network Traffic Filtering (Extended ACL Implementation)

Extended Access Control Lists (ACLs) were deployed at the gateway to enforce granular security policies, such as preventing Guest users from accessing the internal Staff or Server segments while allowing them internet-bound traffic.

```
Router> enable
Router# configure terminal
Router(config)# ip access-list extended COMPANY_VAULT
Router(config-ext-nacl)# deny icmp 192.168.10.0 0.0.0.255 host 192.168.40.20
Router(config-ext-nacl)# deny icmp 192.168.20.0 0.0.0.255 host 192.168.40.10
Router(config-ext-nacl)# deny icmp 192.168.20.0 0.0.0.255 host 192.168.40.20
Router(config-ext-nacl)# deny tcp 192.168.20.0 0.0.0.255 host 192.168.40.20 eq 80
Router(config-ext-nacl)# permit tcp 192.168.10.0 0.0.0.255 host 192.168.40.20 eq 80
Router(config-ext-nacl)# permit tcp 192.168.10.0 0.0.0.255 host 192.168.40.10 eq 80
Router(config-ext-nacl)# permit icmp 192.168.10.0 0.0.0.255 host 192.168.40.10
Router(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 host 192.168.40.10 eq 80
Router(config-ext-nacl)# permit ip host 192.168.30.2 any
Router(config-ext-nacl)# permit ip any any
Router(config-ext-nacl)# exit
Router(config)# end
```

4.6 Network Services (Web and Database Server Hosting)

Web Server :

```
<html>
<head>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
            text-align: center;
            padding: 50px;
        }

        .container {
            background: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0px 0px 10px rgba(0,0,0,0.1);
            display: inline-block;
        }
```

```
        h1 {
            color: #2c3e50;
        }

        p {
            color: #7f8c8d;
        }

        .status {
            color: green;
            font-weight: bold;
        }
    </style>
</head>

<body>
    <div class="container">
        <h1>Mystical Flute Corp</h1>
        <p>Welcome to the Public Web Portal</p>
        <hr>
        <p>Network Status: <span class="status">ONLINE</span></p>
        <p>This site is public and available to all visitors on the network</p>
    </div>
</body>
</html>
```

Database Server:

```
<html>
<head>
    <style>
        body {
            font-family: 'Courier New', monospace;
            background-color: #1a1a1a;
            color: #00ff00;
            text-align: center;
            padding: 50px;
        }

        .vault {
            border: 2px solid #00ff00;
            padding: 20px;
            display: inline-block;
            background-color: #000;
        }

        h1 {
            text-transform: uppercase;
```

```
            letter-spacing: 5px;
        }

        .warning {
            color: red;
            font-weight: bold;
            animation: blink 1s infinite;
        }

        @keyframes blink {
            0% { opacity: 1; }
            50% { opacity: 0; }
            100% { opacity: 1; }
        }
    </style>
</head>
<body>
    <div class="vault">
        <h1>SECURE DATABASE</h1>
        <p class="warning">*** AUTHORIZED ACCESS ONLY ***</p>
        <hr color="#00ff00">
        <p align="left">> Connecting to Core Systems...</p>
        <p align="left">> Authentication Verified: STAFF/ADMIN</p>
        <p align="left">> Log: Customer Data v4.2 Loaded.</p>
        <p align="left">> Log: Financial Records Encrypted.</p>
    </div>
</body>
</html>
```

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Connectivity Testing (ICMP/Ping Analysis)

- Objective : To verify basic network reachability between devices using ICMP (Ping) and confirm that inter-VLAN routing is functioning correctly..
- Ping Test Results

Source Device	Destination Name	Destination IP Address	Result
PC1 (Staff)	Web Server	192.168.40.10	Blocked
PC1 (Staff)	Database Server	192.168.40.20	Blocked
PC-GUEST	Web Server	192.168.40.10	Success
PC-GUEST	Database Server	192.168.40.20	Blocked
PC-ADMIN	Web Server	192.168.40.10	Success
PC-ADMIN	Database Server	192.168.40.20	Success

Table 5.1 Ping Test Results

- Observations: ICMP requests from Staff and Guest devices to the server network are not successful, while the Admin device is able to reach both servers.
- Result: Ping-based reachability is available only for the Admin system. Communication from other VLANs is restricted.
- Conclusion: The network limits ICMP communication to authorized systems, ensuring controlled reachability and reducing exposure of critical servers.

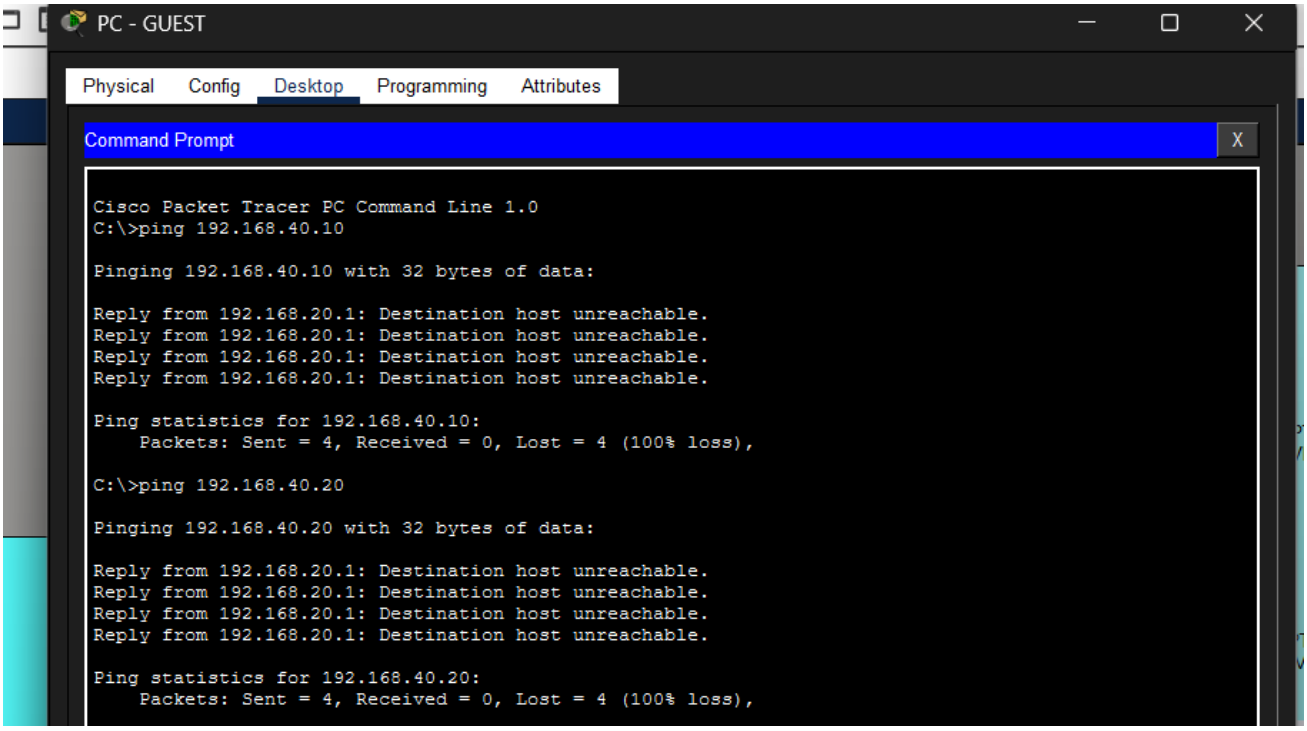


Figure 5.1 ICMP Connectivity Test (End-to-End Ping Analysis)

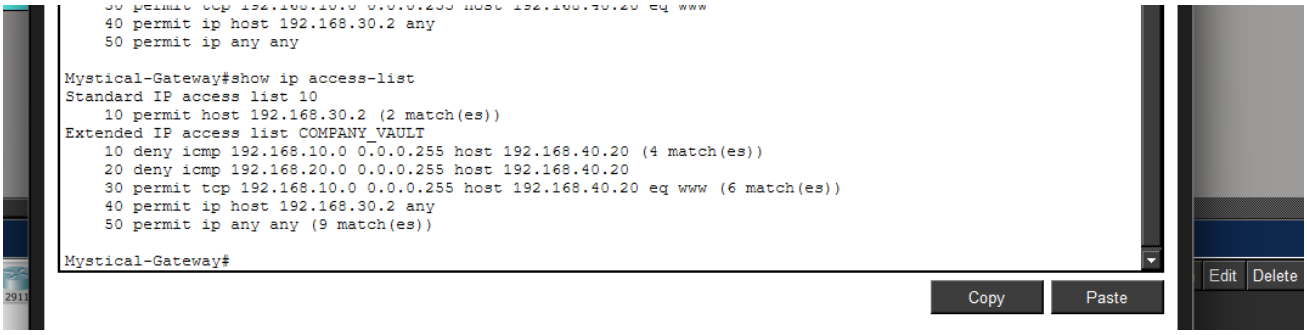
5.2 Security Policy Verification (ACL Testing)

- Objective :To verify that the configured Access Control List (ACL) correctly enforces security policies by allowing only authorized traffic and blocking unauthorized access..
- Test Results :

Source Device	Destination Name	Destination IP Address	Service	Result
PC1 (Staff)	Web Server	192.168.40.10	HTTP	Success
PC1 (Staff)	Database Server	192.168.40.20	HTTP	Success
PC-GUEST	Web Server	192.168.40.10	HTTP	Success
PC-GUEST	Database Server	192.168.40.20	HTTP	Blocked
PC1 (Staff)	Database Server	192.168.40.20	ICMP	Blocked
PC-GUEST	Database Server	192.168.40.20	ICMP	Blocked
PC-ADMIN	All Servers	All Services	HTTP/ICMP	Success

Table 5.2: Access Control List (ACL) Verification Results

- Observation**:** Access to the database server is restricted for Guest devices and ICMP communication is blocked for non-admin users. The Admin device has unrestricted access to all services..
- Result**:** The ACL successfully enforces access restrictions by blocking unauthorized traffic while allowing required communication for administrative purposes.



```
    40 permit ip host 192.168.30.2 any
    50 permit ip any any

Mystical-Gateway#show ip access-list
Standard IP access list 10
    10 permit host 192.168.30.2 (2 match(es))
Extended IP access list COMPANY_VAULT
    10 deny icmp 192.168.10.0 0.0.0.255 host 192.168.40.20 (4 match(es))
    20 deny icmp 192.168.20.0 0.0.0.255 host 192.168.40.20
    30 permit tcp 192.168.10.0 0.0.0.255 host 192.168.40.20 eq www (6 match(es))
    40 permit ip host 192.168.30.2 any
    50 permit ip any any (9 match(es))

Mystical-Gateway#
```


Figure 5.2 Verification of Security Policy via ACL Permit/Deny Logs

5.3 Management Access Audit (SSH Test)

- Objective: To verify secure remote access to the router using SSH and ensure that only authorized devices can establish a connection..
- Test Results:

Source Device	Destination	Result
PC-ADMIN	Router (SSH)	Success
PC1 (Staff)	Router (SSH)	Blocked
PC-GUEST	Router (SSH)	Blocked

Table 5.3 SSH test Results

- Observation: During testing, it was observed that the router does not even present a login prompt to the Staff or Guest devices. When those devices attempt to initiate an SSH session, the router immediately rejects the packet because the source IP address does not match the "Permit" rule in our Management Access Control List. Only the Admin PC, which has the specific IP 192.168.30.2, is granted a secure connection.
- Result: The security configuration effectively limits router access. Only the Admin can access the router's CLI for management. Staff and Guest devices are completely prevented from accessing the router's internal settings, even if they know the administrative credentials. This creates a secure "Management Zone" that is invisible to regular users.

- Conclusion: By using SSH instead of Telnet and applying a VTY Access-Class, we have successfully hardened the management interface. This protects the network from "Unauthorized Configuration" and ensures that the core infrastructure can only be modified by the designated Cybersecurity Administrator.

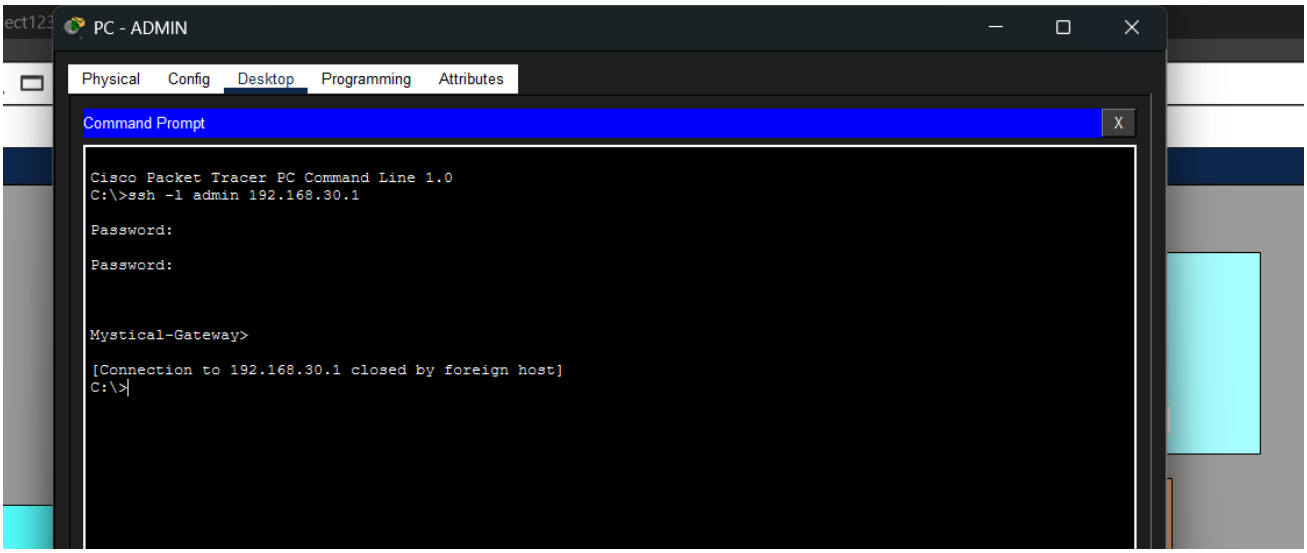


Figure 5.3 Successful Remote Management Access via SSH Audit

5.4 Website Hosting and HTTP Access Test

- Objective: To verify that the web server is successfully hosting the website and that HTTP access is available to authorized users based on the configured network policies.
- Test:

Source Device	Destination Name	Destination IP Address	Result
PC1 (Staff)	Web Server	192.168.40.10	Success
PC1 (Staff)	Database Server	192.168.40.20	Success
PC-GUEST	Web Server	192.168.40.10	Success
PC-GUEST	Database Server	192.168.40.20	Blocked
PC-ADMIN	Web Server	192.168.40.10	Success
PC-ADMIN	Database Server	192.168.40.20	Success

Table 5.4: HTTP Access and Web Hosting Results

- Observation: The web server is accessible through a web browser for authorized users. Staff devices can access both web and database server pages, while Guest devices are restricted from accessing the database server. The Admin device has full access to all hosted services.
- Result: HTTP services are functioning correctly, and access to hosted web pages is controlled according to defined security policies.
- Conclusion: The network successfully delivers web services while restricting access to sensitive resources. This ensures that users can access necessary applications without compromising security.

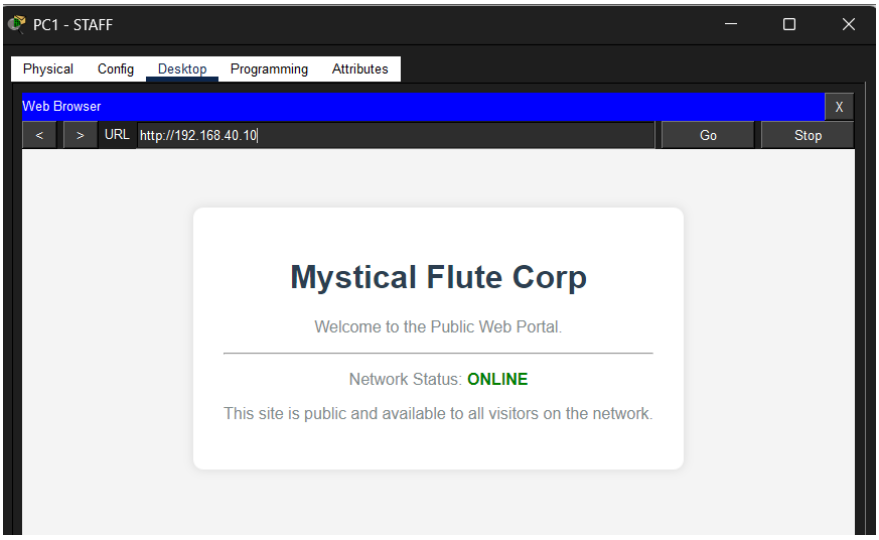


Figure 5.4 Successful HTTP Request to Web Server

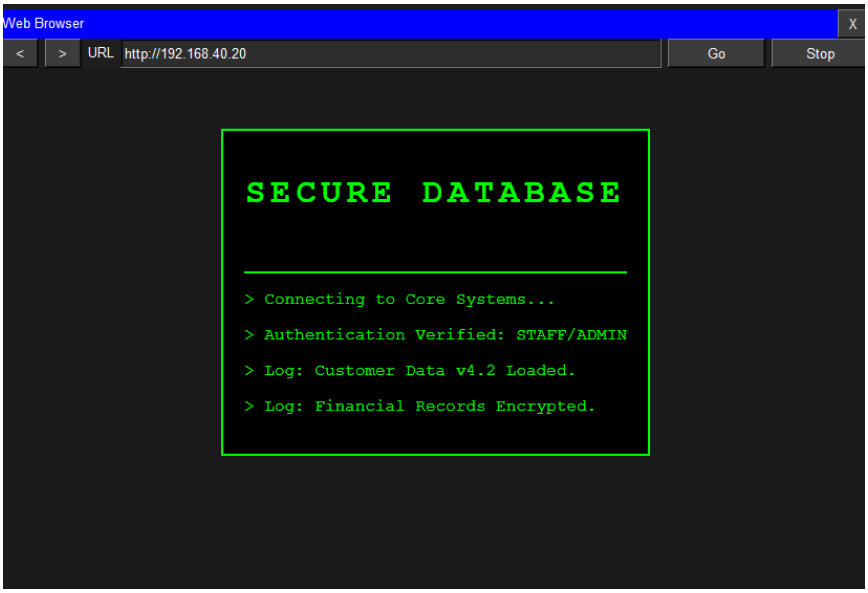


Figure 5.5 Successful Database Query/Connection Test

5.5 Physical Port Security Audit (Layer 2 Hardening)

- Objective: To verify the implementation of MAC-Address Sticky security on the switch and ensure that only authorized hardware can physically access the network. This test confirms that the switch will automatically neutralize any port if an unauthorized device (rogue laptop) is detected.
- Test Results:

Source Device	Port Status	Action Taken	Result
Authorized Staff PC	Secure-Up	Traffic Allowed	Success
Unauthorized Laptop	Error-Disabled	Port Shutdown	Blocked
Authorized Admin PC	Secure-Up	Traffic Allowed	Success

Table 5.5 Port Security Test

- Observation: When the authorized Staff PC is connected to its assigned port, the switch recognizes the "Sticky" MAC address and allows data to flow normally. However, when a different laptop (simulating a hacker) is plugged into that same port, the switch detects a MAC address violation. Within milliseconds, the switch triggers the violation shutdown command, and the port light changes from green to solid red.
- Result: The security implementation ensures that only the original authorized hardware can access the switch port. Any attempt to swap cables or connect a rogue device results in an immediate port shutdown. This prevents "plug-and-play" attacks where an intruder tries to bypass our digital firewalls by gaining physical access to a wall jack.
- Conclusion: The Port Security mechanism effectively hardens the physical layer of our infrastructure. By using the sticky learning method, we have created a self-defending switch that automatically locks down during a security breach, requiring an administrator to manually reset the port after the threat is cleared.

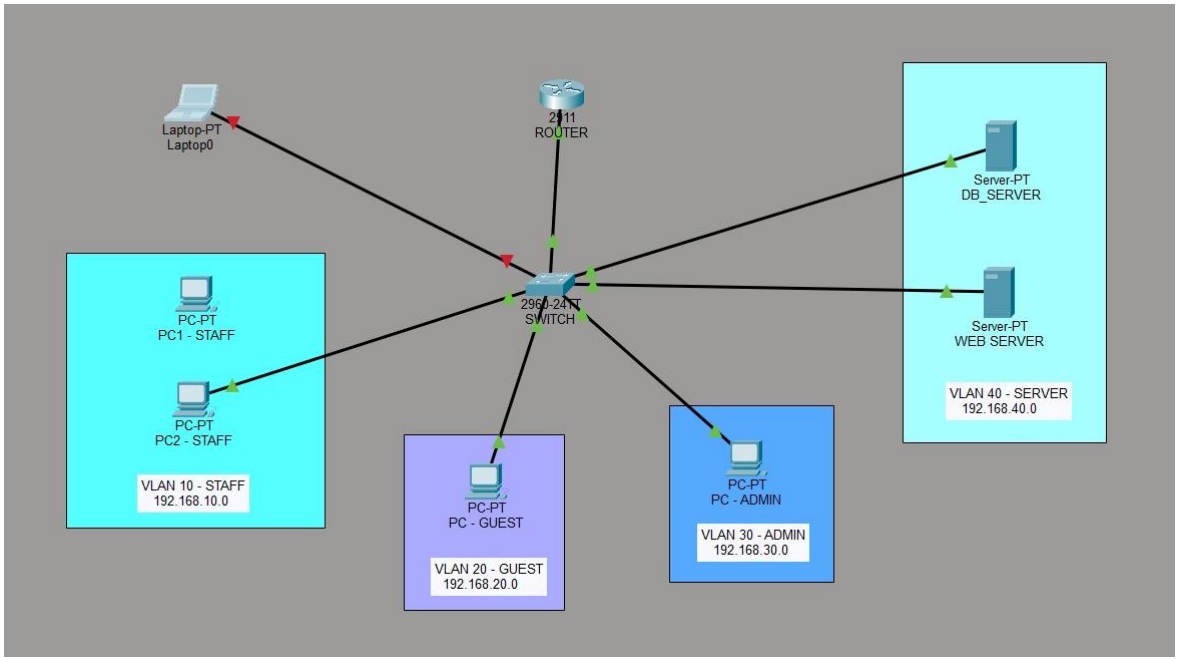


Fig 5.6. Port security

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully achieved the goal of designing and securing a departmental network using Cisco Packet Tracer. By moving away from a traditional flat network and implementing VLAN segmentation, we created a robust environment where traffic is logically isolated and organized.

The implementation of Router-on-a-Stick ensured that different departments could still communicate through a controlled gateway. Furthermore, the addition of Port Security, SSH encryption, and Extended ACLs moved the project from a simple networking task to a security-hardened infrastructure. The final results prove that unauthorized users (Guests) are effectively blocked from sensitive zones (Servers), while authorized users (Admin/Staff) maintain seamless access to necessary resources.

6.2 Future Work

While this project covers the core requirements of network segmentation and access control, it can be expanded in the following ways to meet higher industry standards:

- Dynamic IP Addressing: Implementing a DHCP Server to automatically assign IP addresses to PCs, reducing the manual effort of static configuration.
- Layer 3 Switching: Upgrading from Router-on-a-Stick to a Layer 3 Switch for faster inter-VLAN routing with higher bandwidth and lower latency.
- Redundancy: Implementing Spanning Tree Protocol (STP) or HSRP to ensure the network stays online even if a cable or switch fails.
- Advanced Firewalling: Replacing basic ACLs with a Cisco ASA Firewall to provide stateful packet inspection and deep-level security filtering.
- VPN Integration: Setting up a Virtual Private Network (VPN) to allow the Admin to securely manage the network from an external location

REFERENCES

1. Cisco Systems, Inc. "Cisco Packet Tracer." *Networking Academy*, https://www.netacad.com
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